

GATE-DA

Subject – (Artificial Intelligence)

Mock Test – 1

Topics Covered

Conditional Independence
Bayesian Networks
Variable Elimination
Propositional Logic
Predicate Logic
Reasoning Under Uncertainty
A* Search
Uniform Cost Search
BFS
DFS
Approximate Inference
Heuristic Search Concepts
Minimax
Alpha-Beta Pruning

Section A – Artificial Intelligence

Q1 (1 Mark, MCQ)

DFS starts at node S. S has children A and B (left to right). After popping S and expanding it (left-to-right exploration), what is the exact stack state (top to bottom)?

- A. [A, B] — A on top
- B. [B, A] — B on top
- C. [A, B, S]
- D. [B, A, S]

Q2 (1 Mark, MCQ)

Why does UCS test the goal condition only when a node is POPPED, not when it is inserted into the frontier?

- A. Popping is faster than inserting
- B. A node can be reached via a cheaper path discovered later, after it was first inserted with a higher cost
- C. UCS does not use a goal test at all
- D. Testing on insert would make UCS incomplete

Q3 (1 Mark, MCQ)

BFS is used to check if a graph is bipartite. Starting from node A, A is coloured Red. Its neighbour B is coloured Blue. B's neighbour C is coloured Red. There is an edge between C and A. What does BFS conclude?

- A. Bipartite — C and A are different nodes so no conflict
- B. Not bipartite — C and A are both Red, creating a conflict on edge C—A
- C. Bipartite — C and A are in the same set
- D. Cannot determine without more nodes

Q4 (1 Mark, MCQ)

What is the time complexity of BFS-based bipartite detection on a graph with V vertices and E edges?

- A. $O(V^2)$
- B. $O(V \times E)$
- C. $O(V + E)$
- D. $O(b^d)$ where b is branching factor

Q5 (1 Mark, MCQ)

A node n has $g(n)=12$ (actual cost from start) and $h(n)=8$ (heuristic estimate to goal). What is $f(n)$, and what does it represent?

- A. $f(n)=20$, the estimated total cost of the cheapest solution through n
- B. $f(n)=4$, the remaining distance only
- C. $f(n)=8$, just the heuristic value
- D. $f(n)=12$, just the path cost so far

Q6 (1 Mark, MCQ)

A MIN node has three MAX children with computed values 6, 2, and 9. What value propagates up from this MIN node?

- A. 9 — MIN takes the highest value to be safe
- B. 6 — the first child evaluated
- C. 2 — MIN takes the minimum of [6, 2, 9]
- D. 17 — sum of all children

Q7 (1 Mark, MCQ)

A game tree has $b=4$, $m=6$. Under WORST-CASE move ordering, what is the time complexity of alpha-beta pruning compared to plain minimax?

- A. $O(4^3)$ — alpha-beta always achieves the best case
- B. $O(4^6)$ — identical to plain minimax, since worst ordering causes zero effective pruning
- C. $O(4^6) / 2$ — half the work regardless of ordering
- D. Cannot be computed without knowing the actual leaf values

Q8 (1 Mark, MCQ)

Given: "If the alarm rings, the building is evacuated" ($A \rightarrow E$). We observe the building was NOT evacuated ($\neg E$). What can we validly conclude?

- A. The alarm rang (A) — using Modus Ponens
- B. The alarm did NOT ring ($\neg A$) — using Modus Tollens
- C. Nothing can be concluded
- D. The alarm rang AND the building was evacuated

Q9 (1 Mark, MCQ)

Consider a Bayesian network with nodes $A \rightarrow B \rightarrow C \rightarrow D$, and an additional edge $A \rightarrow D$. Which of the following statements is TRUE?

- A. $A \perp C \mid B$
- B. $A \perp D \mid \{B, C\}$
- C. $B \perp D \mid \{A, C\}$
- D. $A \perp C \mid \{B, D\}$

Q10 (1 Mark, MCQ)

In Variable Elimination on a Bayesian network, you are eliminating variable X which appears in three factors: $f_1(X, A, B)$, $f_2(X, C)$, $f_3(X, D, E)$. After multiplying these factors and summing out X , what is the size of the resulting intermediate factor (in terms of number of variables it's defined over), and what is the maximum number of multiplication operations required to compute the product factor before summing out X (assuming each variable is binary)?

- A. 4 variables $\{A, B, C, D, E\}$; 32 multiplications
- B. 5 variables $\{A, B, C, D, E\}$; 64 multiplications
- C. 5 variables $\{A, B, C, D, E\}$; 32 multiplications
- D. 4 variables $\{A, B, C, D, E\}$; 64 multiplications

Q11 (2 Marks, MCQ)

At a MIN node, child evaluations return values 8, 5, 2 in that order, with incoming $\alpha=4$, $\beta=+\infty$ from the parent. After which child does pruning occur?

- A. After evaluating 8 ($\beta=8$, no prune since $\alpha=4 < \beta=8$)
- B. After evaluating 5: β becomes 5. Check $\alpha(4) \geq \beta(5)$? No. Continue to next child
- C. After evaluating 2: β becomes 2. Check $\alpha(4) \geq \beta(2)$? YES — prune any remaining children
- D. No pruning occurs with these values

Q12 (2 Marks, MCQ)

Minimax computes a root value of 7 for MAX. The actual opponent in a real game makes a mistake and lets MAX achieve a value of 10 instead. Does this violate minimax's optimality guarantee?

- A. Yes — minimax predicted 7 but the actual result was 10, so minimax was wrong
- B. No — minimax guarantees MAX can ACHIEVE AT LEAST 7 against optimal play; against a suboptimal opponent, MAX may do even better
- C. No — minimax always predicts the exact final score regardless of opponent behavior
- D. Yes — minimax is only valid if both players play randomly

Q13 (2 Marks, MCQ)

A heuristic h satisfies $h(n) \leq h^*(n)$ for all n (admissible) but does NOT satisfy $h(n) \leq c(n, n') + h(n')$ for some edge. What is the practical consequence for A*?

- A. A* will return a suboptimal solution
- B. A* remains optimal, but may need to re-open and re-expand some already-explored nodes when a cheaper path is found later
- C. A* becomes incomplete and may never terminate
- D. A* automatically becomes Greedy Best-First Search

Q14 (2 Marks, MCQ)

For a route-finding problem, which of the following heuristics is GUARANTEED to be admissible?

- A. $h(n)$ = straight-line (Euclidean) distance from n to goal
- B. $h(n)$ = $2 \times$ straight-line distance from n to goal
- C. $h(n)$ = actual road distance + 10% buffer
- D. $h(n)$ = number of cities remaining $\times$ average city size

Q15 (2 Marks, MCQ)

A graph has branching factor $b=2$. The optimal solution cost $C^*=20$, and the minimum edge cost in the entire graph is $\epsilon=4$. What is the time complexity bound for UCS?

- A. $O(2^{20})$
- B. $O(2^{(1+\text{floor}(20/4))}) = O(2^6) = 64$
- C. $O(2^4)$
- D. $O(20/4) = 5$, so $O(2^5)$

Q16 (2 Marks, MCQ)

What is the PRIMARY practical limitation of BFS on large search spaces?

- A. Too slow — high time complexity
- B. Must store entire frontier — exponential space
- C. Cannot handle graphs
- D. Requires a heuristic function

Q17 (2 Marks, MCQ)

BFS on tree with $b=3$, goal at $d=4$. Nodes expanded in worst case?

- A. 81
- B. 121
- C. 120
- D. 80

Q18 (2 Marks, MCQ)

Given a Bayesian network: A and B are independent root nodes (no parents), both binary. C has parents A and B. The CPTs are:

$$P(A=1) = 0.3, P(B=1) = 0.6$$

$$P(C=1|A=1,B=1) = 0.9$$

$$P(C=1|A=1,B=0) = 0.5$$

$$P(C=1|A=0,B=1) = 0.4$$

$$P(C=1|A=0,B=0) = 0.1$$

You observe $C=1$. What is $P(A=1 | C=1)$?

- A. 0.300
- B. 0.379
- C. 0.421
- D. 0.450

Q19 (2 Marks, MCQ)

You're running A* search on a graph where the heuristic $h(n)$ is consistent. At some point during the search, node N is popped from the frontier with $f(N) = g(N) + h(N) = 20$. Later, a different node M is popped with $f(M) = 18$. Which of the following is necessarily true?

- A. This cannot happen if h is consistent — f-values must be non-decreasing along the sequence of expansions.
- B. This can happen even with a consistent h, because f-values are only guaranteed non-decreasing along any single path, not across the global expansion order.
- C. This can happen only if h is admissible but not consistent.
- D. This indicates a bug in the implementation regardless of h's properties, since A* always expands nodes in strictly increasing f-order.

Q20 (2 Marks, MCQ)

The converse of $p \rightarrow q$ is:

- A. $\neg p \rightarrow \neg q$
- B. $q \rightarrow p$
- C. $\neg q \rightarrow \neg p$
- D. $p \leftrightarrow q$

Q21 (2 Marks, NAT)

A tree has branching factor $b=3$. The goal node is found at depth $d=3$, and it is the LAST node expanded at that depth (worst case within the level). How many total nodes does BFS expand to find the goal?

Answer: _____

Q22 (2 Marks, NAT)

A weighted graph: $S \rightarrow A$ (cost 3), $S \rightarrow B$ (cost 1), $A \rightarrow G$ (cost 2), $B \rightarrow A$ (cost 1), $B \rightarrow G$ (cost 6). What is the minimum cost from S to G?

Answer: _____

Q23 (2 Marks, NAT)

A node has $g(n)=18$ and the true remaining cost is $h^*(n)=7$. The given heuristic is $h(n)=9$. By how much does $h(n)$ overestimate $h^*(n)$?

Answer: _____

Q24 (2 Marks, NAT)

A MAX root has 2 MIN children. Left MIN child has leaves [5, 8, 3]. Right MIN child has leaves [7, 2, 9]. What is the minimax value at the root?

Answer: _____

Q25 (2 Marks, MSQ)

Which of the following statements about A* and UCS are TRUE? (Select ALL that apply.)

- A. A* with $h(n)=0$ for all n behaves identically to UCS
- B. UCS is a special case of A* where the heuristic always equals the true remaining cost
- C. If $h(n)$ is consistent, A* never needs to re-open an already-explored node
- D. A* always expands strictly fewer nodes than UCS, for any admissible heuristic

Q26 (2 Marks, MSQ)

Which of the following statements about Alpha-Beta pruning are TRUE? (Select ALL that apply.)

- A. Alpha-beta pruning can return a different (but faster-computed) root value compared to plain minimax in some cases
- B. The pruning condition for cutoff is $\alpha \geq \beta$
- C. Alpha-beta achieves its best-case time complexity of $O(b^{(m/2)})$ regardless of move ordering
- D. Pruned subtrees are never visited or evaluated at all

Q27 (2 Marks, NAT)

A game tree has branching factor $b=8$ and maximum depth $m=6$. Under perfect move ordering, alpha-beta examines approximately $b^{(m/2)}$ nodes. Compute this value.

Answer: _____

Q28 (2 Marks, NAT)

Consider the Bayesian network $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ (a simple chain, all binary variables). You want to compute $P(E)$ using Variable Elimination, eliminating variables in the order A, B, C, D. At each elimination step, a new factor is generated over the remaining variables connected to the eliminated variable in the current factor set.

Assuming all variables are binary, what is the total number of multiplication operations performed across the entire elimination order (sum over all four elimination steps)? Give your answer as an integer.

Answer: _____

Q29 (2 Marks, MCQ)

Let X, Y, Z, W be random variables. Suppose the joint distribution satisfies $X \perp Y \mid Z$ and $X \perp W \mid Z$, but no other independence or conditional independence holds among any other subset unless implied by these two. Which of the following **MUST** hold given only this information?

- A. $X \perp \{Y, W\} \mid Z$
- B. $X \perp Y \mid \{Z, W\}$
- C. $Y \perp W \mid Z$
- D. $X \perp (Y, W) \mid Z$, assuming $Y \perp W \mid \{X, Z\}$ also holds (i.e., apply the intersection property)

Q30 (2 Marks, NAT)

In a Bayesian network used for approximate inference via rejection sampling, suppose you want to estimate $P(Q=1 \mid E=1)$ where Q and E are both binary nodes in the network, and $P(E=1) = 0.08$. You generate $N = 5000$ samples from the prior joint distribution using forward sampling. In expectation, how many samples will be rejected (i.e., not match the evidence $E=1$)?

Answer: _____

Section B – Aptitude

Q31 (1 Mark, MCQ)

A train travels at a constant speed and crosses a 240 m platform in 24 seconds and crosses a 120 m platform in 18 seconds. What is the speed of the train?

- A. 10 m/s
- B. 15 m/s
- C. 20 m/s
- D. 25 m/s

Q32 (1 Mark, MCQ)

Choose the word that is most nearly opposite in meaning to “PARSIMONIOUS”:

- A. Frugal
- B. Lavish
- C. Cautious
- D. Retacent

Q33 (1 Mark, MCQ)

If “CODING” is written as “DPEJOH”, how is “FLOWER” written in the same code?

- A. GMPXFS
- B. GMPXES
- C. FMPXFS
- D. GNPXFS

Q34 (2 Marks, MCQ)

A and B can complete a work in 12 days. B and C can complete the same work in 15 days. A and C can complete it in 20 days. In how many days can A, B, and C complete the work together?

- A. 8
- B. 10
- C. 9
- D. 7.5

Q35 (2 Marks, MCQ)

Study the statements:

All researchers are readers. Some readers are writers.

Which conclusion necessarily follows?

- A. Some researchers are writers
- B. Some writers are researchers
- C. No valid conclusion follows
- D. All writers are readers

Q36 (2 Marks, MCQ)

The ratio of the present ages of P and Q is 3:5. After 9 years, the ratio becomes 3:4. What is the present age of P?

- A. 9
- B. 12
- C. 15
- D. 18

Q37 (2 Marks, MCQ)

A cube of side 4 cm is painted on all faces and then cut into 1 cm cubes. How many small cubes have exactly two faces painted?

- A. 16
- B. 24
- C. 8
- D. 32

Q38 (2 Marks, MCQ)

In a row of children facing north, Reema is 7th from the left and Sanjay is 9th from the right. If they interchange positions, Reema becomes 12th from the left. How many children are there in the row?

- A. 18
- B. 19
- C. 20
- D. 21

Q39 (2 Marks, MCQ)

Select the option that best completes the analogy:

Thermometer : Temperature :: Barometer : ?

- A. Wind speed
- B. Humidity
- C. Atmospheric pressure
- D. Altitude

Q40 (2 Marks, MCQ)

A sum of money becomes 1.5 times itself in 4 years at simple interest. What is the rate of interest per annum?

- A. 10.5%
- B. 12.5%
- C. 11%
- D. 12%